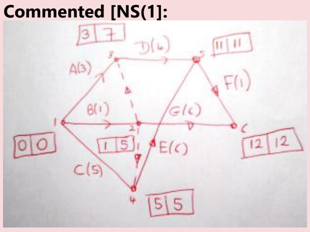
Discrete Mathematics: Activity on arc revision

1 The table below shows the activities that need to be completed as part of a task. The immediate predecessors and durations of each activity are shown.

Activity	Immediate predecessors	Duration (hours)
A	-	3
B	-	1
C	-	5
D	A, B	4
E	B, C	6
F	D, E	1
G	B	6

(a) Represent the above on an activity-on-arc diagram.

Commented [NS(1]:



(b) Find the early event time and late event time for each event.

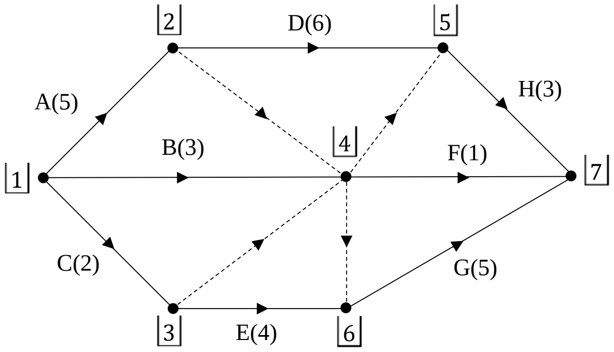
(c) Give the minimum completion time and list the critical activities.

Commented [NS(2]: Minimum completion time is 12. The critical activities are C, E, F.

(d) Find the float of activity G.

Commented [NS(3]: 12-1-6=5

2 The activity on-arc diagram below shows the eight activities that need to be completed as part of a project. The duration of each activity is shown in brackets. The events are labelled $\boxed{1}$, $\boxed{2}$, $\boxed{3}$, ...



(a) Produce a table showing the immediate predecessors of each activity.

Commented [NS(4]: A, B, C no predecessors
D has A
E has C
F has A, B, C
G has A, B, E (NOT C)
H has B, C, D (NOT A)

(b) Perform a forwards and backwards pass to find the early and late times of each event. Write down the minimum completion time and list the critical activities.

Commented [NS(5]: In the order of event numbering: (0,0), (5,5), (2,5), (5,9), (11,11), (6,9), (14,14).

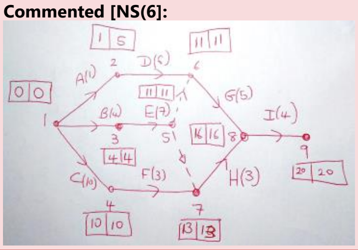
So minimum completion time 14 and critical activities: A, D, H

3 The table below shows the activities that need to be completed as part of a task. The immediate predecessors and durations of each activity are shown.

Activity	Immediate predecessors	Duration (hours)
A	-	1
B	-	4
C	-	10
D	A	6
E	B	7
F	C	3
G	D, E	5
H	E, F	3
I	G, H	4

(a) Represent the above on an activity-on-arc diagram.

Commented [NS(6]:



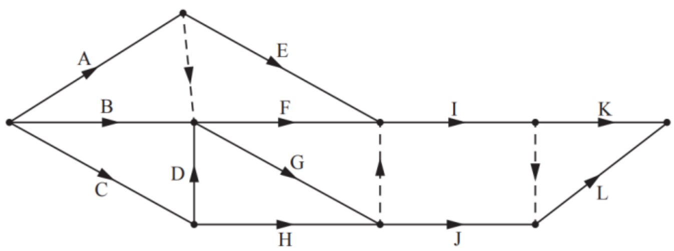
(b) Find the early event time and late event time for each event.

(c) Find the float of each activity, and hence identify the critical activities.

Commented [NS(7]: A has float 5-0-1=4
B has float 4-0-4=0
C has float 10-0-10=0
D has float 11-1-6=4
E has float 11-4-7=0
F has float 13-10-3=0
G has float 16-11-5=0
H has float 16-13-3=0
I has float 20-16-4=0

So every activity is critical except for A and D

4 The network shown below represents a project using activity on arc.



(a) Explain the reason for the dummy on the left of the diagram.

Commented [NS(8]: F depends on A and B, but E depends only on A

It is given that activity E is a critical activity. The manager of this project therefore claims that activities A, E, I, and K must form a critical path for the network.

(b) Is the manager's claim correct? Explain your reasoning.

Commented [NS(9]: Not necessarily. A and I must be critical (if either had float, then so would E. BUT L could be critical instead of (or as well as) K.

The durations of the activities are shown below:

Activity	A	B	C	D	E	F	G	H	I	J	K	L
Duration, hours	7	4	6	2	10	6	3	7	5	6	5	3

(c) Carry out a forward pass and a backward pass through the activity network, showing the early event time and late event time at each vertex of the network.

Commented [NS(10]: Not having the vertices numbered is an irritation here, but a deliberate act on the part of the writer, to make you think about precedence. There are only 2 ways to number it and the times (in number order) are (0,0), (7,7), (6,10), (8,11), (11,17), (17,17), (22,22), (22,24), (27,27). The second and third of these could be the other way around.